

Decision Tree Task: Disease Diagnosis

Objective:

Help students understand how to use a Decision Tree Classifier to predict diseases based on patient symptoms and age.

Training Dataset:

Fever	Cough	Fatigue	Headache	Age	Disease
Yes	Yes	Yes	No	25	Flu
Yes	Yes	No	Yes	32	COVID-19
No	Yes	No	No	45	Common Cold
Yes	No	Yes	Yes	60	Typhoid
No	No	No	No	22	Healthy
Yes	Yes	Yes	Yes	35	COVID-19
No	Yes	Yes	No	29	Common Cold
Yes	No	Yes	No	50	Typhoid
No	No	Yes	No	40	Healthy
Yes	Yes	No	Yes	28	Flu

Task:

A new patient arrives with the following symptoms:

- Fever: Yes
- Cough: Yes
- Fatigue: Yes
- Headache: No
- Age: 30

 Using a Decision Tree Classifier trained on the above dataset, predict the most likely disease for this patient.

Instructions:

1. Train a Decision Tree Classifier using the dataset provided.
2. Visualize the decision tree to understand the decision paths.
3. Predict the disease for the patient based on the given input.
4. Justify your prediction using the tree rules.
5. Optional: Try this using Orange ML or Python (sklearn).

Expected Output:

- The predicted disease (e.g., Flu, COVID-19, etc.)
- Explanation of the prediction path based on the decision tree.